Open-Source AI Models Survey

Multimodal Support, Architecture, Parameter Size, and Resource Estimation

Objective

Survey recent open-source AI models by analyzing:

- Architecture
- Parameter count
- Multimodal capabilities
- Supported input types
- Estimated CPU RAM and GPU VRAM requirements
- Quantized variants (FP16, INT8, INT4/Q4)
- Quantization impact on memory usage

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1. Important Concepts

What Is a Model Parameter?

A parameter is a learned weight inside a neural network.

Example:

- 7B model = 7 billion parameters
- 70B model = 70 billion parameters

Larger parameter counts generally:

- improve reasoning capability
- improve knowledge retention
- increase hardware requirements
- increase inference latency

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2. Memory Estimation Formula

FP16 (16-bit)

Each parameter uses:

```
2 bytes
```

Approximate memory:

```
VRAM = Parameters × 2 bytes
```

Example:

```
7B × 2 bytes ≈ 14 GB
```

INT8 Quantization

Each parameter uses:

```
1 byte
```

Memory becomes approximately:

```
50% of FP16
```

INT4 / Q4 Quantization

Each parameter uses:

```
0.5 bytes
```

Memory becomes approximately:

```
25% of FP16
```

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3. Types of AI Models

Model Type	Purpose
LLM	Text generation and reasoning
Multimodal LLM	Text + image/audio/video understanding
Embedding Model	Convert text/images into vectors
Speech Model	Audio processing
Vision Model	Image understanding

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4. Survey of Recent Open-Source AI Models

A. Meta Llama 3.1

Attribute	Value
Organization	Meta
Architecture	Transformer Decoder
Modalities	Text
Variants	8B, 70B, 405B
Context Length	128K
Open Source	Yes

Multimodal Support

- Native multimodal: No
- Text only

Resource Estimation

Variant	FP16 VRAM	INT8 VRAM	INT4/Q4 VRAM
8B	~16 GB	~8 GB	~4-5 GB

Variant	FP16 VRAM	INT8 VRAM	INT4/Q4 VRAM
70B	~140 GB	~70 GB	~35-40 GB
405B	~810 GB	~405 GB	~200 GB

Common Usage

- General chatbots
- RAG systems
- Agents
- Coding assistants

B. Mistral Large / Mixtral

Attribute	Value
Organization	Mistral AI
Architecture	Mixture of Experts (MoE)
Modalities	Text
Variants	7B, 8x7B, 8x22B
Open Source	Yes

Architecture Insight

MoE activates only selected experts during inference.

Benefits:

- lower inference cost
- faster execution
- better scalability

Multimodal Support

- Native multimodal: No
- Text only

Resource Estimation

Variant	FP16 VRAM	INT8 VRAM	INT4/Q4 VRAM
7B	~14 GB	~7 GB	~4 GB

Variant	FP16 VRAM	INT8 VRAM	INT4/Q4 VRAM
8x7B	~90 GB	~45 GB	~24 GB
8x22B	~280 GB	~140 GB	~70 GB

C. Qwen 2.5 VL

Attribute	Value
Organization	Alibaba
Architecture	Vision-Language Transformer
Modalities	Text + Image + Video
Variants	7B, 72B
Open Source	Yes

Multimodal Capabilities

Supports:

- text
- image understanding
- video understanding
- OCR
- chart analysis
- document understanding

Resource Estimation

Variant	FP16 VRAM	INT8 VRAM	INT4/Q4 VRAM
7B	~16 GB	~8 GB	~4-5 GB
72B	~145 GB	~72 GB	~36 GB

Real-World Use Cases

- multimodal assistants
- document AI
- image question answering
- video summarization

D. LLaVA

Attribute	Value
Organization	University of Wisconsin
Architecture	LLM + Vision Encoder
Modalities	Text + Image
Base Models	Vicuna/Llama
Open Source	Yes

Architecture

Combines:

- CLIP vision encoder
- language model

Resource Estimation

Variant	FP16 VRAM	INT8 VRAM	INT4/Q4 VRAM
7B	~16 GB	~8 GB	~4 GB
13B	~26 GB	~13 GB	~7 GB
34B	~70 GB	~35 GB	~18 GB

Multimodal Features

Supports:

- image captioning
- visual reasoning
- chart understanding
- screenshot analysis

E. Phi-3 Vision

Attribute	Value
Organization	Microsoft
Architecture	Small Language Model + Vision

Attribute	Value
Modalities	Text + Image
Variants	4B
Open Source	Yes

Key Strength

Optimized for smaller hardware.

Resource Estimation

Variant	FP16 VRAM	INT8 VRAM	INT4/Q4 VRAM
4B	~8 GB	~4 GB	~2-3 GB

Usage

- edge devices
- laptops
- lightweight multimodal apps

F. Whisper

Attribute	Value
Organization	OpenAI
Architecture	Encoder-Decoder Transformer
Modalities	Audio
Variants	Tiny, Base, Small, Medium, Large
Open Source	Yes

Capabilities

- speech recognition
- translation
- transcription

Resource Estimation

Variant	FP16 VRAM	INT8 VRAM	INT4/Q4 VRAM
Tiny	~1 GB	~0.5 GB	~0.25 GB
Base	~1.5 GB	~0.8 GB	~0.4 GB
Medium	~5 GB	~2.5 GB	~1.3 GB
Large	~10 GB	~5 GB	~2.5 GB

G. CLIP

Attribute	Value
Organization	OpenAI
Architecture	Dual Encoder
Modalities	Text + Image
Open Source	Yes

Purpose

CLIP generates aligned embeddings for:

- text
- images

Used heavily in:

- image search
- vector databases
- multimodal retrieval

Resource Estimation

Variant	FP16 VRAM	INT8 VRAM	INT4/Q4 VRAM
ViT-B/32	~1-2 GB	~1 GB	~0.5 GB
ViT-L/14	~4-5 GB	~2 GB	~1 GB

H. Sentence Transformers

Attribute	Value
Organization	UKPLab
Architecture	Siamese Transformer
Modalities	Text
Open Source	Yes

Purpose

Convert text into embeddings.

Used for:

- semantic search
- RAG systems
- similarity matching
- clustering

Popular Models

Model	Parameters
all-MiniLM-L6-v2	~22M
all-mpnet-base-v2	~109M
bge-large-en	~335M

Resource Estimation

Model	FP16 VRAM	INT8 VRAM	INT4/Q4 VRAM
MiniLM	~100 MB	~50 MB	~25 MB
MPNet	~500 MB	~250 MB	~125 MB
BGE Large	~1.5 GB	~0.8 GB	~0.4 GB

I. Stable Diffusion XL

Attribute	Value
Organization	Stability AI

Attribute	Value
Architecture	Diffusion Model
Modalities	Text → Image
Open Source	Yes

Capabilities

- image generation
- image editing
- inpainting
- style transfer

Resource Estimation

Variant	FP16 VRAM	INT8 VRAM	INT4/Q4 VRAM
SDXL Base	~12 GB	~6 GB	~3 GB

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5. Quantization Comparison

What Is Quantization?

Quantization reduces numerical precision of model weights.

Example:

Precision	Bytes Per Parameter
FP32	4 bytes
FP16	2 bytes
INT8	1 byte
INT4/Q4	0.5 byte

Why Quantization Matters

Benefits:

- lower VRAM usage

• lower RAM usage
• faster inference
• enables local deployment

Tradeoffs:

• slight accuracy loss
• possible reasoning degradation
• reduced numerical precision

Quantization Example

Llama 3.1 8B

Precision	Estimated VRAM
FP16	~16 GB
INT8	~8 GB
INT4/Q4	~4-5 GB

Observation

INT4 reduces memory usage by approximately 75% compared to FP16.

This allows:

• consumer GPU deployment
• laptop inference
• edge AI systems

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6. CPU RAM vs GPU VRAM

GPU VRAM

Required when model runs on GPU.

Impacts:

• inference speed
• parallel processing
• batch size

CPU RAM

Required when:

- model partially offloaded
- running on CPU
- loading tokenizer/cache

Typically:

```
CPU RAM requirement ≈ 1.5x to 2x VRAM requirement
```

Example:

Model	GPU VRAM	CPU RAM
7B FP16	~16 GB	~24-32 GB
70B INT4	~40 GB	~64-80 GB

7. Multimodal Model Comparison

Model	Text	Image	Audio	Video
Llama 3.1	Yes	No	No	No
Mixtral	Yes	No	No	No
Qwen 2.5 VL	Yes	Yes	No	Yes
LLaVA	Yes	Yes	No	No
Phi-3 Vision	Yes	Yes	No	No
Whisper	No	No	Yes	No
CLIP	Yes	Yes	No	No
SDXL	Text Prompt	Generates Image	No	No

8. Key Observations

Large Models

Advantages:

- stronger reasoning
- better instruction following
- larger knowledge capacity

Disadvantages:

- massive hardware requirements
- slower inference
- expensive deployment

Small Models

Advantages:

- edge deployment
- lower latency
- cheaper inference

Disadvantages:

- weaker reasoning
- reduced context understanding

Quantized Models

Advantages:

- local deployment possible
- reduced memory consumption
- faster inference

Disadvantages:

- slight quality degradation
- lower precision arithmetic

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9. Real-World Production Considerations

Consumer GPU Deployment

GPU	Suitable Models
RTX 3060 12GB	7B INT4
RTX 4090 24GB	13B FP16 / 70B INT4
A100 80GB	70B FP16
H100	405B distributed inference

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10. Common AI Stack in Modern Systems

Typical RAG Pipeline

```
User Query
     ↓
Embedding Model
     ↓
Vector DB (FAISS/ChromaDB)
     ↓
Retriever
     ↓
LLM
     ↓
Final Response
```

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11. Conclusion

Recent open-source AI systems increasingly support multimodal capabilities including text, image, audio, and video understanding.

Model size directly impacts:

- GPU VRAM requirements
- CPU RAM requirements
- inference latency
- deployment cost

Quantization techniques such as INT8 and INT4 significantly reduce memory usage, enabling deployment on consumer hardware while introducing only moderate quality degradation.

Smaller efficient models are becoming increasingly important for edge AI and low-cost inference, while large multimodal systems dominate advanced reasoning and enterprise-scale AI applications.

Modern AI systems commonly combine:

- embedding models
- vector databases
- multimodal LLMs
- quantized inference
- retrieval pipelines

forming the foundation of current production AI architectures.